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Performance evaluation for a novel optoelectronic device for noninvasive monitoring thrombosis

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ABSTRACT

Deep vein thrombosis (DVT) is the main type of venous thromboembolism, which is the most common thrombotic disease. In this study, we developed a set of non-invasive thrombosis diagnostic device. A series of experiments were employed to evaluate its performance including stability, robustness, sensitivity and test on human. The results showed that near infrared spectroscopy (NIRS) could reliably monitor thrombosis for DVT patients. With its noninvasive, rapid, continuous, and real-time features, NIRS can detect human hemodynamic parameters and show advantages in DVT diagnosis.

1. Introduction

Thrombosis is the most common one of the three major cardiovascular diseases worldwide [1,2]. Thrombus can be formed in any section of the vein system, otherwise the most frequent part is the deep veins of leg [3]. Most DVT patients are at risk for complications such as ulcers, skin changes and chronic swelling [4]. However, the embolization of the thrombus in the lungs can be fatal, which is the main concern [3]. Furthermore, the incidence of the disease may recur [5,6]. Therefore, the diagnosis of DVT is of great importance. There are many detection methods; nevertheless, it remains challenges to improve them on convenience, accuracy and sensitivity. The swelling degree and skin temperature method is not a quantitative analysis, meanwhile delayed. The ankle brachial index (ABI) method, the most widely used means, is simple and noninvasive, but not accurate [7,8]. The ultrasonic imaging is not suitable for continuous detection over a long time [9]. The magnetic resonance imaging (MRI) method is expensive with low microvascular imaging quality [10].

In order to detect thrombosis effectively, near infrared spectroscopy (NIRS) has been prompted as a potential method to diagnose venous thrombosis, especially DVT [11–13]. Based on the different absorptions with variant near-infrared wavelengths of chromophores and ingredients, NIRS is a noninvasive continuous detection for oxygenated and deoxygenated hemoglobin measurement [14–19].

Some researchers used NIRS for hemodynamic detection on brain, and also on capillary, arteriole and veinlet thrombosis in calf [20]. For example, Helmelt et al. employed the blood model experiment to test the potential of NIRS on DVT detection [12]. Scott et al. conducted the cuff experiment to analyze the difference between normal legs and those of thrombotic patients using NIRS [11]. Similarly, the groups of Hosoi and Yamaki reported the thrombosis diagnostic method with ambulatory venous retention index (AVRI) on the condition of movement paradigm [4,21]. The studies above have promoted the research of NIRS in the field of thrombus detection. However, there are relatively few studies devoted to performance evaluation of these detection methods with NIRS. The performance of these detection methods remains to be explored.

In this study, we developed a novel noninvasive NIRS device for DVT diagnosis. In order to evaluate the stability, robustness, sensitivity and test on human, we performed a series of corresponding experiments (temperature test, pressure test, electromagnetic interference test, dark noise interference test, liquid-model optical experiment, and cuff experiment). The results showed good stability, robustness, sensitivity and test on human for DVT monitoring.

2. Methods and experiment design

2.1. Device design

This noninvasive optoelectronic device for thrombosis consisted of probe, functional module and computer as shown in Fig. 1b. We designed a six-channel NIRS probe with one LED source and six sensors. Due to the uneven surface of the calf, we used a flexible printed circuit board to fit avoiding external light interference. We chose the

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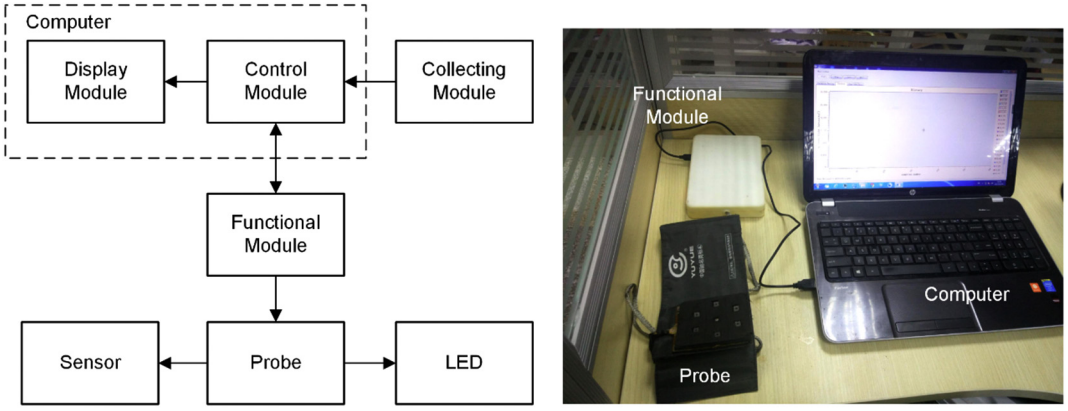


Fig. 1. Device design. (a) Structure chart; (b) Real products.

L735.805.850-40C32P LED (EPITEX Company, at 735 nm, 805 nm and 850 nm wavelengths) as the light source, which was widely accepted for hemodynamic detection with NIRS [22]. The OPT101 (TI Company) was the sensor, which sensitively responded to the selected three light wavelengths [23].

We shaped the black flexible plastic as the welded probe circuit, then imbedded it on the probe circuit [13]. The source-sensor distance was 2.8 cm as shown in Fig. 2a, b. The angle between two adjacent sensors and the light source was 60°. Fig. 1a showed that the functional module converted optical signals into digital signals. Then it transmitted the digital signals from the probe to the software in the computer. The computer received the collected hemodynamic parameters from the collecting module and displayed the relative hemodynamic graph. The software was developed by LabWindows/CVI (NI Company).

2.2. Evaluated performance parameters

In the design of this device, stability, robustness, sensitivity and clinical utility are considered. The performance of this device is evaluated by several factors below.

For stability, we need to monitor the trend of the light intensity output voltage over a long time. Notably, the received optical signals are relatively weak [13]. Hence, the light source and the sensor, which may generate heat, easily suffer interference. In terms of robustness, external interference deserves consideration. Additionally, the obtained data are easy to interfered by predictable noise, such as dark noise and background light noise [24]. It is also valuable to explore whether or not the system is interfered by them. Electromagnetic interference should also be considered as an important factor for weak biological signals.

When considering sensitivity, we can observe the real-time output of monitoring results through the variable composition of analyte. Thus, the corresponding dynamic changes of analyte may imply real-time monitoring and sensitivity. In order to analyze the clinical utility of the system, we also need a simulation experiment for prolonged period.

2.3. Experiment design

For comprehensive evaluations of performance, we designed five experiments in three ways. All of these experiments were performed with an ambient temperature at 25 °C. The stability test was temperature test. The robustness concluded pressure test, electromagnetic

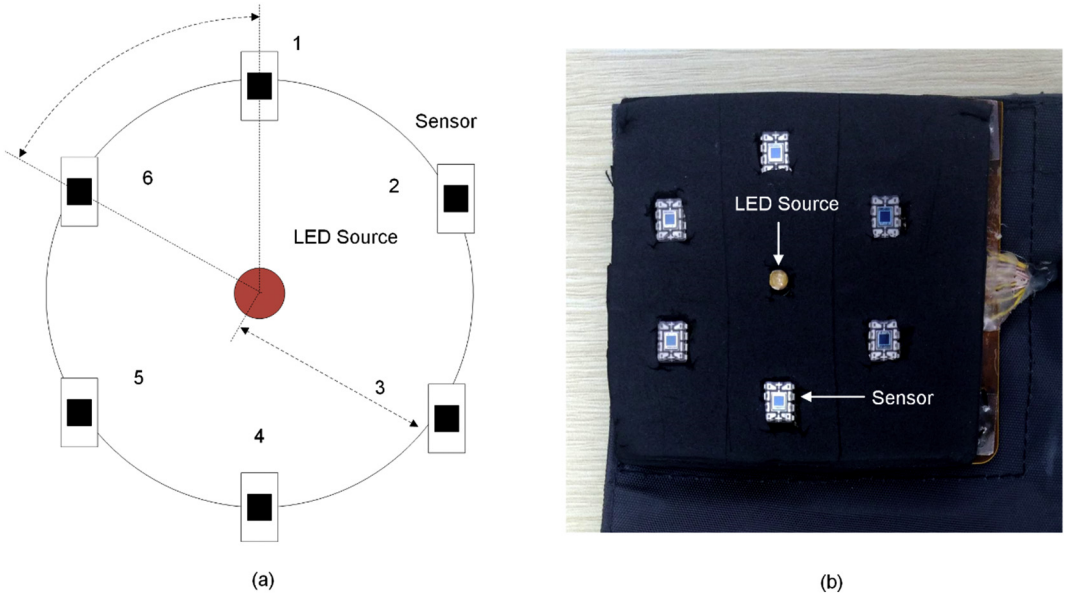


Fig. 2. Probe. (a) Structure of source and sensor; (b) Real products.

interference test and the dark noise test. The liquid-model optical experiment represented sensitivity. The cuff experiment indicated the performance of testing on human. All the data were analyzed by MATLAB (2010b, Mathworks).

2.3.1. Stability with temperature test

Since the probe was tied to the calf, the tightness might cause external pressure to the sensor. It was essential to analyze the influence of the external pressure. We conducted this test in a closed room. We attached the probe to a standard tissue-like phantom and wrapped both of them with a black cloth to avoid external light. The system ran for 8 h simulating the working condition of thrombus monitoring. The light intensity voltage data were recorded. Light tended to generate heat. Therefore, it was necessary to monitor temperature of the light source and the sensor. Meanwhile, the temperature of the light source and sensor were collected with a wireless thermometer every 10 min. The trend of the voltage and the temperature of the source and the sensor could help us determine the stability of the system for a long period of time.

2.3.2. Robustness with pressure test, electromagnetic interference test and dark noise test

We made advantage of the solid phantom to do pressure test. We ran the system and set no pressure on the probe in the first 60 s. 0.5 kg weight was added up to 1.5 kg on the back of the probe every 60 s, which increased the external pressure. All the weights were taken off in the 240th second. The monitoring data were collected during 270 s.

The electromagnetic interference test was conducted by a mobile phone (iPhone 7 plus, Apple Company). The mobile phone was placed 10 cm near the probe. When test started, the system ran for the first minute without electromagnetic interference. Then we had been making a call to a fixed-line phone (HCD868 TSD, TCL Company) on the whole second minute. Afterwards, we repeated these steps above to the sixth minutes. Hence, data of five and a half minutes were collected when the system worked with or without the electromagnetic interference. The test presented whether the electromagnetic interference had an impact on the system. For quantitative analysis, we also brought the two sets of data (one set of no interfered data and another set of interfered data) into the noise percent equation as following:

$$NP_i \text{ (noise percent)} = \frac{U_{no\ interfered} - U_{interfered}}{U_{interfered}} \times 100\%. \quad (1)$$

In Eq. (1), the $U_{no\ interfered}$ was the average data during no interfered time, while the $U_{interfered}$ was the average data during interfered period. NP_i denoted the noise percent of the electromagnetic interference.

In the dark noise test, we recorded the data for 60 min with background light off. For quantitative analysis, we also collected for another 60 min with light on. Two sets of the data preceding were brought into the noise percent formula as below:

$$NP_d \text{ (noise percent)} = \frac{U_{on} - U_{off}}{U_{off}} \times 100\%. \quad (2)$$

According to the equation above, the U_{on} were the collected data with light on. In contrast, the U_{off} were the collected data with light off. NP_d represented the noise effect by the background light.

2.3.3. Sensitivity with liquid-model optical experiment

This experiment was performed with all lights off in the optical dark room. We stuck the probe to the polyethylene container with black cloth wrapped and placed the stirrer into the container. We set it on a magnetic stirrer console and poured 600 mL of water into the container. The magnetic stirrer started to run. The diluted carbon black ink was injected into the container every minute as shown in Fig. 3. The data was collected and the curve was drawn for 6 min till the end. The liquid-model optical experiment could accurately show that whether the variation of the light absorption of the solution made the real-time sensitive output.



Fig. 3. Liquid-model optical experiment design.

2.3.4. Test on human with cuff experiment

The volunteer remained normal sitting position in a quiet room without light interference. We wrapped the upper arm of the volunteer with the sphygmomanometer belt as normal measure method. The upper arm was placed on the desktop with palm up as shown in Fig. 4. The increasing pressure to 140 mm Hg made the arm blood circulation blocked for three minutes. Later, we quickly released the pressure valve and kept for two minutes. Simultaneously, the software recorded the relative changes of oxygenated hemoglobin, deoxygenated hemoglobin and total hemoglobin [25]. The cuff experiment presented whether the system could monitor effectively before or after simulated thrombotic symptoms. The system could show accurate results and proper determination of the hemodynamics variation from thrombosis without any other interference.

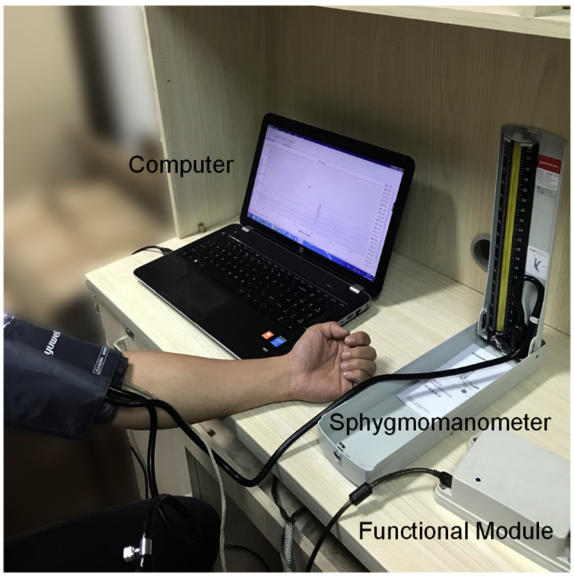


Fig. 4. Cuff experimental design.

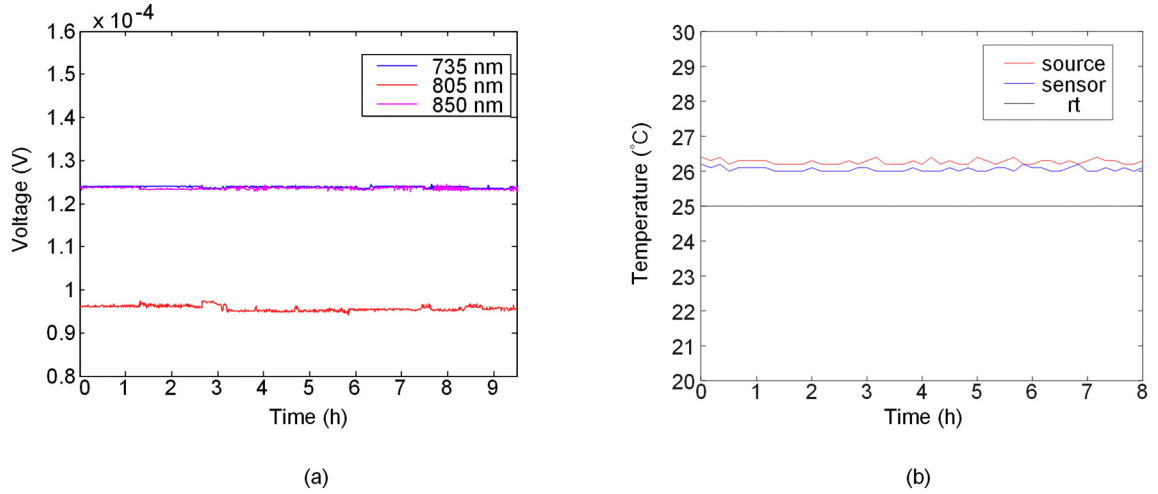


Fig. 5. Temperature test results. (a) Voltage of three wavelengths; (b) Temperature of the light source and sensor. rt: room temperature.

3. Results

The temperature test results in Fig. 5b showed that the temperature of the source was around 26.3 °C, while the sensor was around 26.2 °C. All the data stayed stable with slight fluctuation but no major trends. The output voltage data of the three wavelengths in Fig. 5a were steady during the lasting 8 h.

The pressure test results are shown in Fig. 6. The light intensity voltage data of three wavelengths were collected by the sensor in a certain channel. We figured out that even the weight increased 0.5 kg in the 60th, 120th, 180th second every time, the received intensity voltage data did not change much. At the 240th second, we took all the weight off. The light intensity voltage stayed stable.

Fig. 7 showed the output light intensity in channel 1. We could figure out that electromagnetic interference did not change any intensity output in the two interfered periods (as dash area). The output curve of the three wavelengths were almost steady during the whole 270 s. Similar results were found in channel two to channel six. We also listed the noise percent of interference in Table 1. All the NP_i values were less than 1. Subsequently, the electromagnetic interference hardly affected the system operation.

As can be seen from Fig. 8, the dark noise was about 2550 μV with ± 150 μV fluctuation. Meanwhile, we listed the NP value of the three wavelengths from six channels. All the NP values were below 1 in Table 2. Therefore, the influence from dark noise could be ignored.

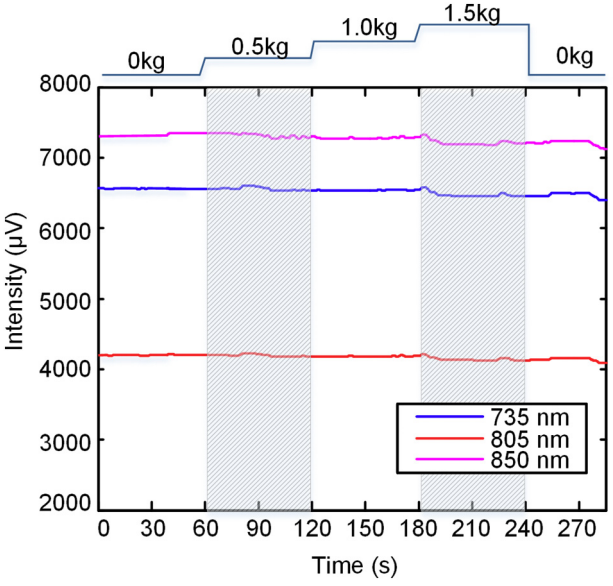


Fig. 6. Pressure test results.

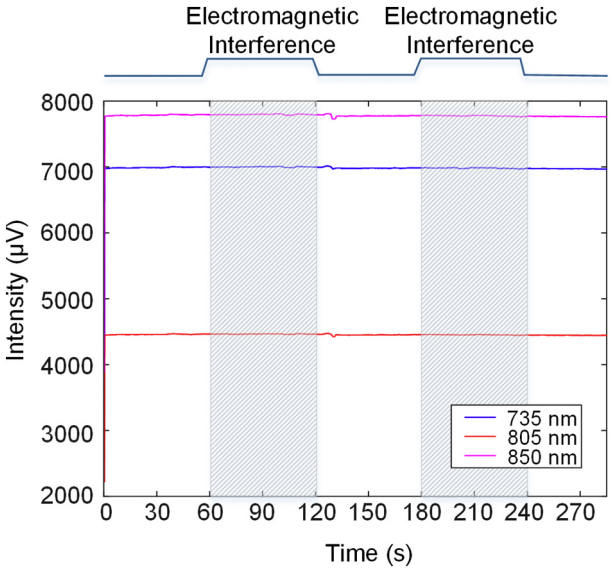


Fig. 7. Electromagnetic Interference test results in channel 1.

Table 1
Noise percent for six channels in electromagnetic interference test.

Wavelength	735 nm	805 nm	850 nm
NP_i			
Channel			
1	0.08	0.10	0.09
2	0.13	0.15	0.14
3	0.01	0.11	0.02
4	0.03	0.14	0.03
5	0.11	0.12	0.13
6	0.42	0.44	0.44

At the end of the liquid-model optical experiment, we used MATLAB to analyze the collected data. The light intensity received by the sensors were shown in Fig. 9. The diluted ink was constantly injected into the container as time went on. The light intensity voltage shaped a step-down trend. Moreover, three wavelengths performed similar features.

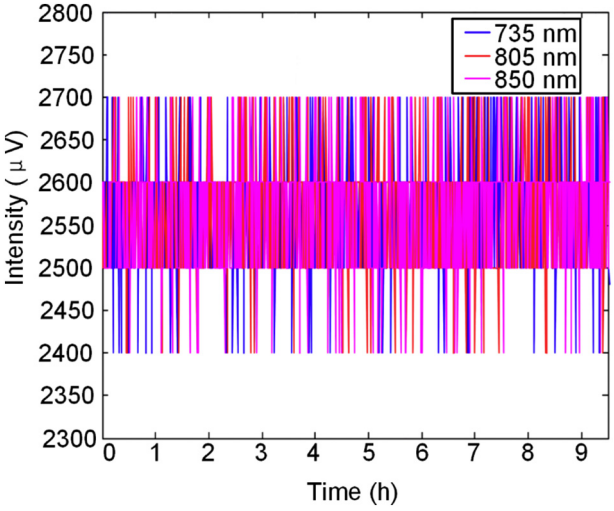


Fig. 8. Dark noise test results.

Table 2
Noise percent for six channels in dark noise test.

Wavelength	735 nm	805 nm	850 nm
NP_d			
Channel			
1	0.25	0.04	0.37
2	0.14	0.3	0.1
3	0.15	0.1	0.25
4	0.02	0.11	0.3
5	0.18	0.05	0.12
6	0.45	0.6	0.05

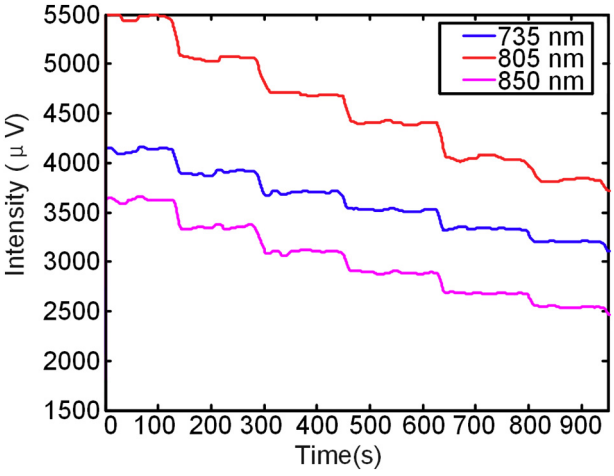


Fig. 9. Liquid-model optical experiment results.

In the cuff experiment, hemodynamic parameters were relatively stable in the first 50 s as shown in Fig. 10. As the pressure increased to 140 mm Hg, the arm blood circulation was blocked. The concentration of oxygenated hemoglobin (oxy-Hb) declined gradually, simultaneously the concentration of deoxygenated hemoglobin (deoxy-Hb) slowly rose. The concentration of total hemoglobin remained relatively stable. After releasing the pressure valve, the concentration of oxy-Hb rose rapidly beyond the resting level. It came to stabilize afterwards. The concentration of deoxy-Hb had the contrast performance. The total content of hemoglobin increased slightly, followed by the same trend toward the initial level. From a physiological point of view, microscopic changes in hemoglobin in human tissues were interrelated.

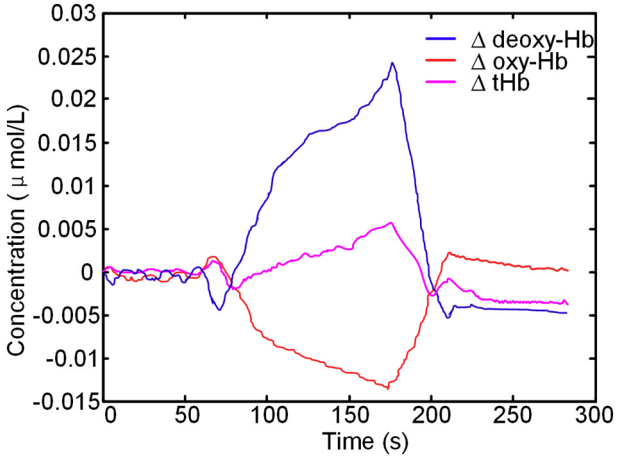


Fig. 10. Cuff experiment results. tHb: total hemoglobin.

4. Summary and discussion

In this study, we performed a series of reliability tests on the performance of NIRS-based thrombus monitors. Previous studies have not specifically tested the reliability of NIRS monitoring for thrombosis. The light source, the sensors, the circuits and other parts in the system were susceptible to interference, resulting in performance reduced. For these impossible factors, we carried on the long-time test for the stability. A variety of interferences concluded pressure from external forces on the sensor, electromagnetic interference and dark noise. We employed relevant experiments to verify the robustness with interference above. We also conducted a liquid-model optical experiment to analyze the sensitivity. Finally, the cuff experiment simulating thrombosis was used to test the actual performance on human. The comprehensive tests above demonstrated the reliability of the system.

In the actual performance evaluation, we detected the output and the temperature of the source and the sensor. They were stable by running the system up to 8 h, which testified the stability of the system. The pressure test was performed to verify that the tightness of the probe bond had no effect on the system. While the system was on, we dialed the mobile phone, finding that electromagnetic interference did not work. During the 3 h dark noise test, the NP_d of three wavelengths were less than 1%, which also indicated that the background light failed to affect. Therefore, the system performed good robustness. In the liquid-model optical experiment, the variation of the signal received by the sensor was in synchronization with the injection the diluted ink, which showed good sensitivity of the system. Meanwhile, the received light attenuation signal had a linear relationship with the solution concentration, which meant good linearity of the system. According to the cuff experiment, the collected data curve was smooth. The system at the same time detected the changes of human tissue oxygen metabolism. As a result, the system performed well on human.

In vivo test, we only used the cuff experiment to simulate human thrombotic symptoms. The sample data was limited. In the future, we will conduct clinical trials for thrombosis in hospital situation with this system. Collecting a big pool of sample data will more efficiently improve the reliability, sensitivity, and specificity of the system.

In summary, we conducted a comprehensive performance evaluation of this noninvasive optoelectronic device for thrombosis. The system has good performance in terms of stability, robustness, sensitivity and test on human. Later, we will make more improvements on the portability of the probe and analysis capability of the instrument. Through the collection and analysis of big data samples, we can anticipate thrombosis prediction on normal people. We expect that this

system can further help more people stay away from thrombotic diseases and maintain their health.

Acknowledgments

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